

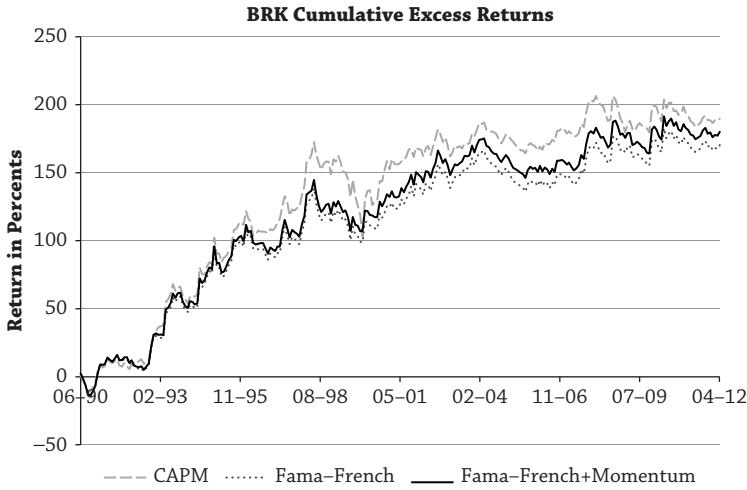


Figure 10.2

The high R^2 indicates the fit of the CAPM benchmark is very good compared to the typical fit for a CAPM regression for an individual stock.

The excess return generated by Berkshire Hathaway relative to the CAPM risk-adjusted benchmark is

$$\begin{aligned} r^{ex} &= r_i - r^{bmk} \\ &= r_i - (0.49r_f + 0.51r_m). \end{aligned}$$

The average of the excess returns is the alpha, $\alpha = E(r^{ex}) = 0.72\%$ per month. Figure 10.2 plots the cumulative excess returns of Berkshire Hathaway. The solid line corresponds to the CAPM benchmark. Note that the slope of the cumulative excess returns flattens in the more recent period, especially since the mid-2000s. In 2012, Berkshire Hathaway has a capitalization over \$220 billion. In the early 1990s, Berkshire's market capitalization was less than \$10 billion. As Berkshire Hathaway has grown bigger, its average excess returns have shrunk. Buffett himself said in 2011,

The bountiful years, we want to emphasize, will never return. The huge sums of capital we currently manage eliminate any chance of exceptional performance. We will strive, however, for better-than-average results and feel it fair for you to hold us to that standard.⁸

Even Berkshire Hathaway is subject to the law of decreasing returns to scale.

Size and Value-Growth Benchmarks

Eugene Fama and Kenneth French, two of the most influential scholars in finance, introduced a benchmark in 1993 that extended the CAPM to include factors that

⁸ Berkshire Hathaway's 2010 shareholder letter.